



NAME:- ARYAN SINGH SIKARWAR

REG NO:- 24BCE0403

EXPERIMENT NO:- 1

DATE:- 7-JAN-2026

MPMC LAB

Aim (Statement)

To write and simulate an **8051 Assembly Language Program (ALP)** using **Keil μ Vision** to perform **16-bit arithmetic operations** such as **addition, subtraction, multiplication, and division** on two numbers, and to verify the results using memory/registers.

Apparatus Required

1. Computer / Laptop
- 2. Keil μ Vision IDE**
- 3. 8051 Microcontroller** (e.g., AT89C51 – for reference/simulation)
4. Windows Operating System
5. Lab Manual / Experiment Instructions

Question 1

Aim

To perform **8-bit addition, subtraction, multiplication, and division** using 8051 ALP.

Algorithm

1. Start
2. Load two 8-bit numbers
3. Perform addition and store result
4. Perform subtraction and store result
5. Perform multiplication and store result
6. Perform division and store quotient & remainder
7. Stop

Flow Chart

Start



Load two 8-bit numbers



Addition → Store



Subtraction → Store



Multiplication → Store



Division → Store



Stop

Question 2

Aim

To add multiple 8-bit numbers and observe the **Carry (CY) flag** using Keil simulator.

Algorithm

1. Start
2. Load first data into accumulator
3. Add next data one by one
4. Monitor CY flag after each addition
5. Store final sum
6. Stop

Flow Chart

Start



Load first data



Add next data



Check CY flag



Repeat addition



Store result



Stop

Question 3

Aim

To calculate $y = x^2 + 2x + 9$ using lookup table in 8051.

Algorithm

1. Start
2. Load x value (0–9) in R0
3. Read x^2 from lookup table (200H)
4. Calculate $2x$
5. Add $x^2 + 2x + 9$
6. Store result in R2
7. Stop

Flow Chart

Start



Load x in R0



Get x^2 from LUT



Calculate $2x$



Add $x^2 + 2x + 9$



Store y in R2



Stop

Question 4

Aim

To display the string “**VIT university**” using 8051 ALP.

Algorithm

1. Start
2. Load address of string
3. Read character
4. Display character
5. Check for end of string
6. Stop

Flow Chart

Start



Load string address



Read character



Display character



End of string?

↓ No → Repeat

↓ Yes

Stop

Question 5

Aim

To perform **16-bit addition, subtraction, multiplication, and division** using 8051 ALP.

Algorithm

1. Start
2. Load two 16-bit numbers
3. Perform 16-bit addition
4. Perform 16-bit subtraction
5. Perform multiplication
6. Perform division
7. Store all results

8. Stop

Flow Chart

Start



Load two 16-bit numbers



Addition → Store



Subtraction → Store



Multiplication → Store



Division → Store



Stop

CODE 1:-

```

# Code 1 :-
⇒ ORG 0000H

MOV A, #25H
MOV RO, #05H

ADD A, RO
MOV 30H, A

MOV A, #25H
CLR C
SUBB A, RO
MOV 31H, A

MOV A, #25H
MOV B, RO
MUL AB
MOV 32H, A
MOV 33H, B

MOV A, #25H
MOV B, RO
DIV AB
MOV 34H, A
MOV 35H, B

HERE : SJMP HERE
END

```

CODE 2:-

Code 2 :-

```
=> ORG 0000H
    MOV A, # 92H ;
    MOV B, # 23H ;
    ADD A, B
    JNC L1
    INC R0
L1 : MOV B, A
    MOV A, # 66H
    ADD A, B
    JNC L2
    INC R0
L2 : MOV B, A
    MOV A, # 87H
    ADD A, B
    JNC L3
    INC R0
L3 : MOV B, A
    MOV A, # 0F5H
    ADD A, B
    JNC L4
    INC R0
L4 :
    END
```

CODE 3:-

#	Code 3 :-	
⇒	ORG 000H	
	MOV DPTR, #200H	
	MOV A, #03H	
	MOV R1, A	
	MOV R0, A	
	ADDC A, R1	
	MOV R1, A	
	MOV A, R0	
	MOVC A, @A + DPTR	
	ADDC A, R1	
	ADDC A, #09H	
	ADDC A, R1	
	MOV R2, A	
	HERE : SJMP HERE	
	ORG 200H	
	DB 00H, 01H, 04H, 09H, 10H, 19H,	
	24H, 31H, 40H, 51H	
	END	

CODE 4:-

Code 4 :-

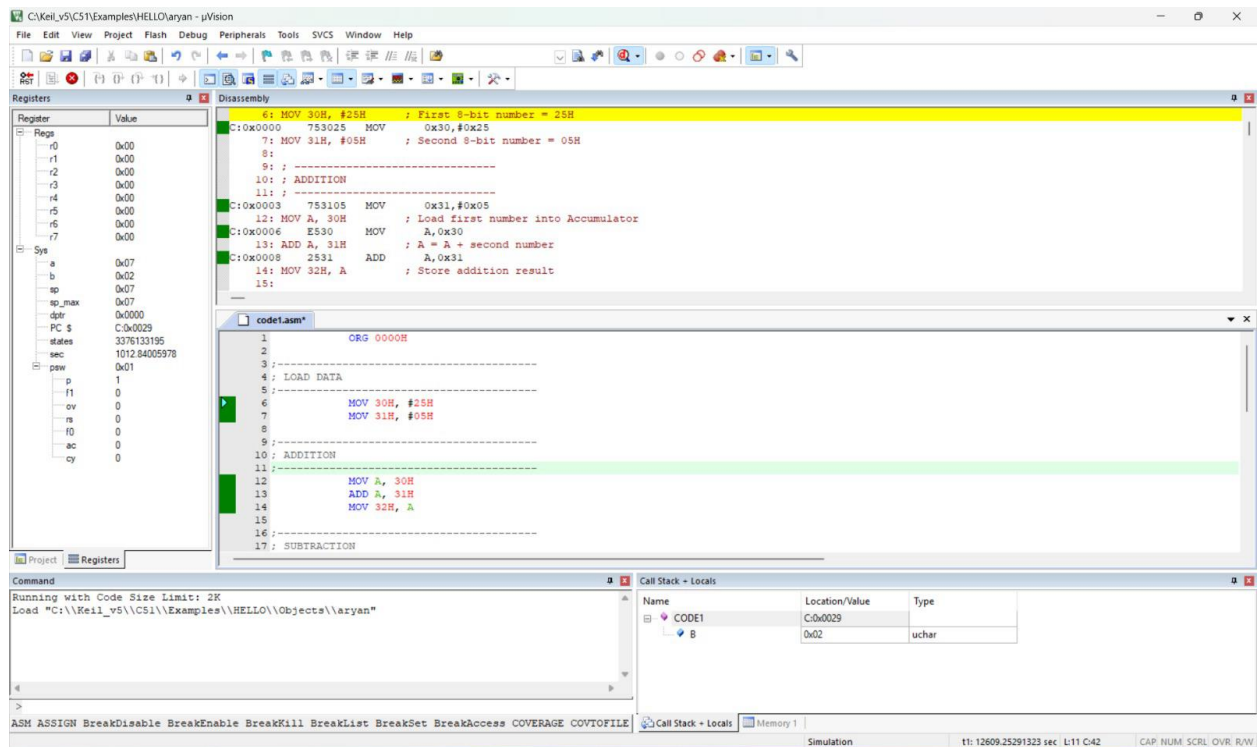
```
=> ORG 000H
    MOV A, # 00H
    MOV DPTR, # 200H
    MOV R1, # 0EH
    MOV R0, # 40H
    LOOP : CLR A
          MOVC A, @A + DPTR
          MOV @R0, A
          INC DPTR
          INC R0
          DJNZ R1, LOOP
    HERE : SJMP HERE
    ORG 200H
    DB " VIT University "
    END
```

CODE 5:-

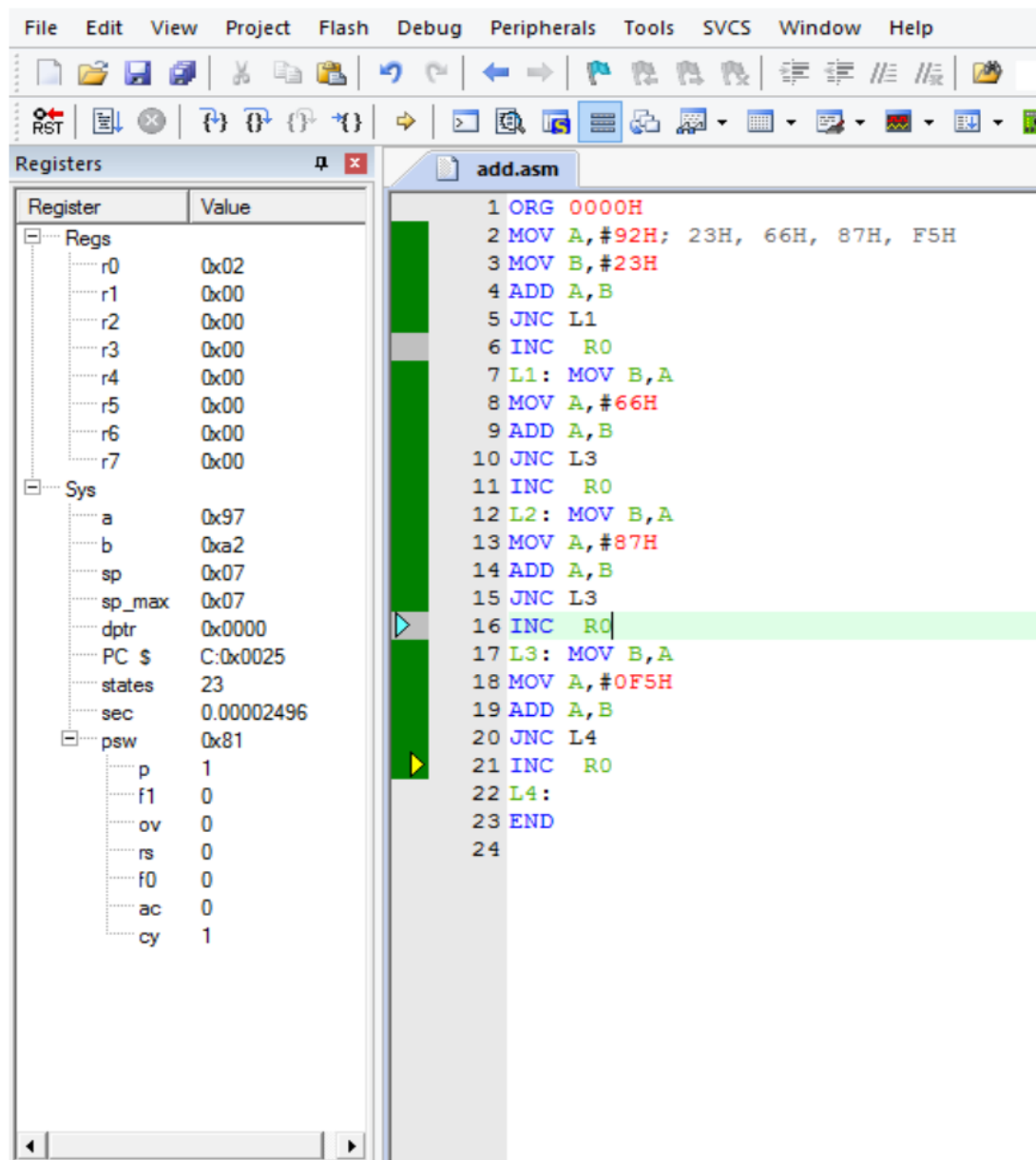
#	Code 5 :-	
	ORG 0000H	
	MOV 30H, #12H	
	MOV 31H, #34H	
	MOV 32H, #00H	
	MOV 33H, #04H	
	MOV A, 31H	
	MOD A, 32H	
	MOV 35H, A	
	MOV A, 30H	
	ADDC A, 32H	
	MOV 34H, A	
	MOV A, 31H	
	CLR C	
	SUBB A, 32H	
	MOV 37H, A	
	MOV A, 30H	
	SUBB A, 32H	
	MOV 36H, A	
	MOV A, 31H	
	MOV B, 33H	
	MUL AB	
	MOV 3BH, A	
	MOV 3AH, B	
	HERE : SJMP HERE	
	END	

SIMULATION IN KEIL SOFTWARE :-

CODE 1:-



CODE 2:-



CODE 3:-

File
Edit
View
Project
Flash
Debug
Peripherals
Tools
SVCS
Window
Help

Registers

Register	Value
Regs	
r0	0x03
r1	0x06
r2	0x18
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x18
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0200
PC \$	C:0x000F
states	187766253
sec	203.73942383
psw	0x00

xsquare2.asm

```

1  ORG 000H
2  MOV DPTR, #200H
3  MOV A, #03H
4  MOV R1, A
5  MOV R0, A
6  ADDC A, R1
7  MOV R1, A
8  MOV A, R0
9  MOVC A, @A+DPTR
10 ADDC A, #09H
11 ADDC A, R1
12 MOV R2, A
13 HERE: SJMP HERE
14
15 ORG 200H
16 DB 00H, 01H, 04H, 09H, 10H, 19H, 24H, 31H, 40H, 51H
17 END
18

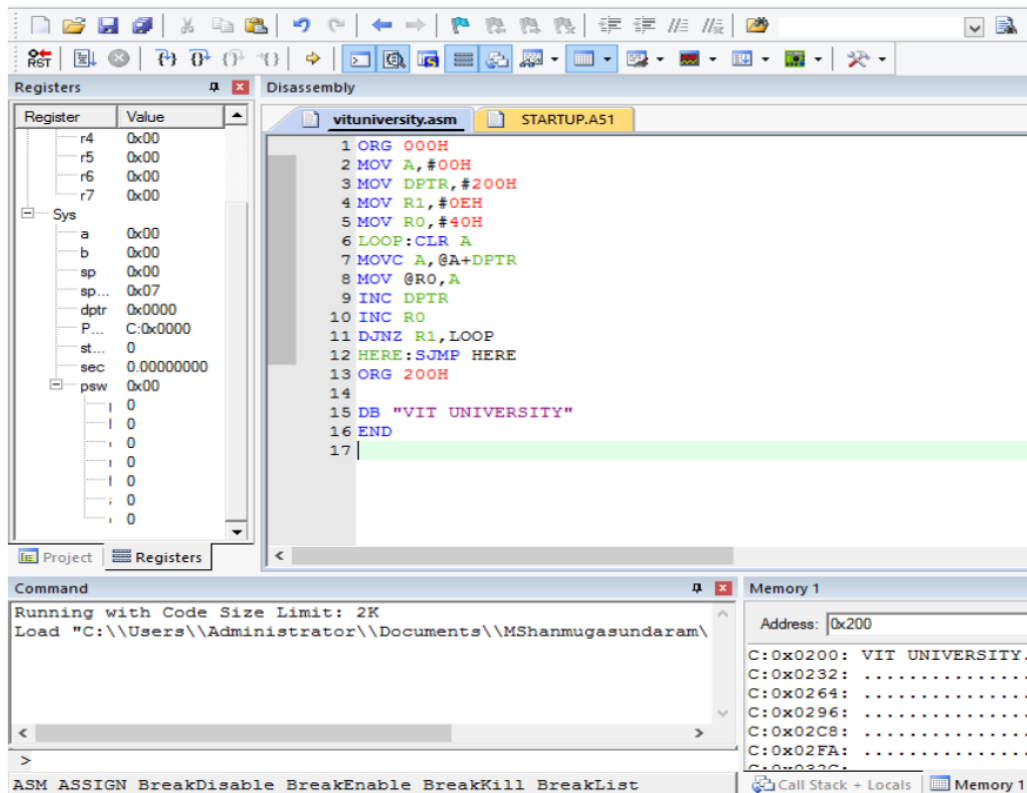
```

Memory 2

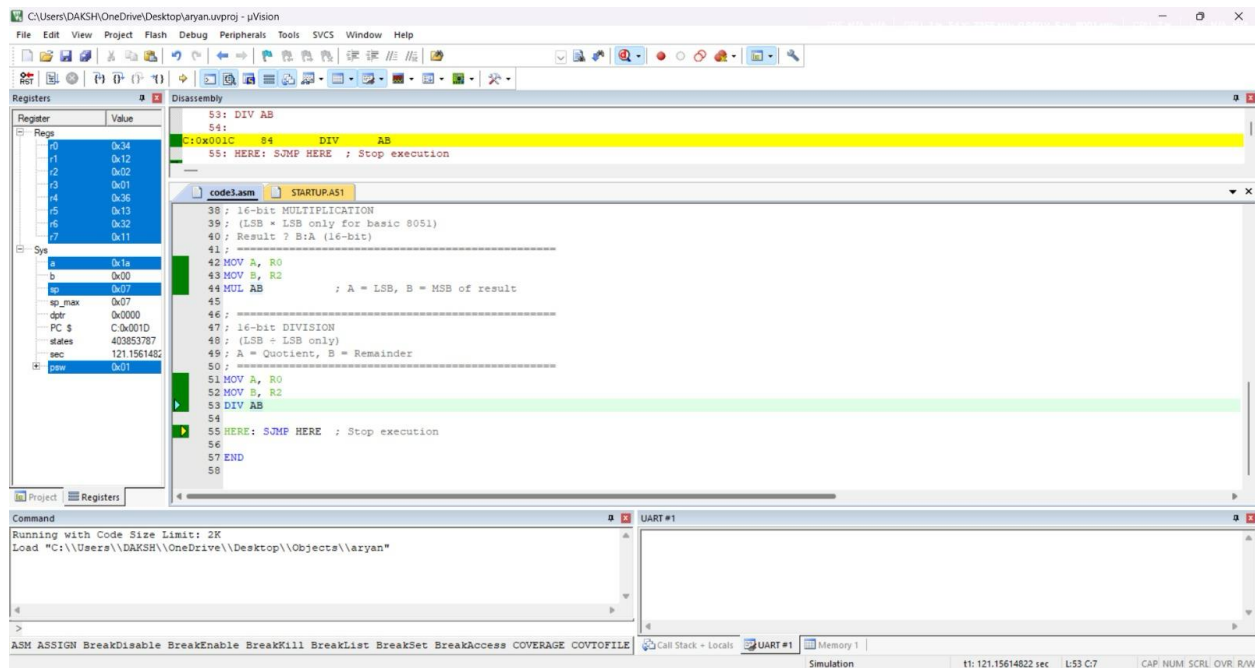
Address: d:000H

D:0x00:	03	06	18	00	00	00	00	00	00	00	00	00	00	00	00
D:0x1E:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
D:0x3C:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00
D:0x5A:	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00

CODE 4:-



CODE 5:-



Observation / Inference / Results

Question 1

The 8051 ALP for **8-bit addition, subtraction, multiplication, and division** was successfully executed in **Keil μVision**. The results were correctly obtained in the specified registers and memory locations. The multiplication result appeared in registers **A and B**, and the division operation produced correct **quotient and remainder** values.

Question 2

The addition of the given data values was successfully simulated. The **Carry (CY) flag** was set whenever the sum exceeded 8 bits. The Keil simulator clearly displayed the CY flag status after each addition, confirming correct overflow detection.

Question 3

The program correctly computed the value of $y = x^2 + 2x + 9$ using a lookup table stored in code memory. By changing the value of x (**0–9**) in the simulator, the corresponding output y was correctly obtained in register **R2**, verifying proper lookup table access and arithmetic operations.

Question 4

The string “**VIT university**” was successfully displayed character by character using the 8051 ALP. The program correctly accessed the string from program memory and terminated upon encountering the null character.

Question 5

The 8051 ALP for **16-bit arithmetic operations** was successfully executed. The results of **addition and subtraction** were accurate, and the **multiplication and division** operations produced correct outputs within the limitations of the 8051 instruction set.

Final Result

All the given programs were **successfully written, assembled, simulated, and verified** using **Keil μ Vision**, and the observed results matched the expected theoretical values.